

IT HOME LAB

Documentation & Setup Guide

Executive Summary

This document details the end-to-end construction of a professional IT home lab built to simulate enterprise-level IT infrastructure. The lab was designed to develop and demonstrate practical skills directly aligned with Level 1 and Level 2 IT Support roles.

The environment was built entirely on VirtualBox using Windows Server 2022 as the domain controller and Windows 11 Pro as the client machine. All configurations were performed hands-on, including Active Directory management, DNS and DHCP services, Group Policy enforcement, VPN access, and security hardening.

Skills demonstrated include user and group administration, network troubleshooting, remote access configuration, security event monitoring, and real-world helpdesk ticket resolution — all of which directly reflect the responsibilities listed in the target IT Support role.

Platform	Domain	Client OS	Duration
VirtualBox + Win Server 2022	lab.local	Windows 11 Pro	May 2026

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May 2026

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1. Lab Overview & Objectives

1.1 Purpose & Objectives

This home lab was constructed to develop and validate practical IT support competencies in a controlled, virtualized environment. The primary objective was to replicate the infrastructure commonly found in small-to-medium enterprise environments, enabling hands-on experience with the tools and technologies encountered in Level 1 and Level 2 IT support roles.

The lab directly addresses the following job description requirements:

- Deliver Level 1 & Level 2 IT support via remote tools
- Troubleshoot Windows 10/11, desktop applications, hardware issues, and basic networking
- Log, track, and resolve incidents using ticketing systems while meeting SLA targets
- Create, manage, and support user accounts in Active Directory and Azure AD
- Support Microsoft 365 services and hybrid infrastructure operations
- Configure and maintain VPN connections for secure remote access
- Monitor and manage backup solutions and Hyper-V virtual machines

1.2 Lab Architecture

The lab runs entirely within VirtualBox on a single host machine, using a NAT network to provide internet access while maintaining an isolated private network. The following diagram illustrates the complete network topology:

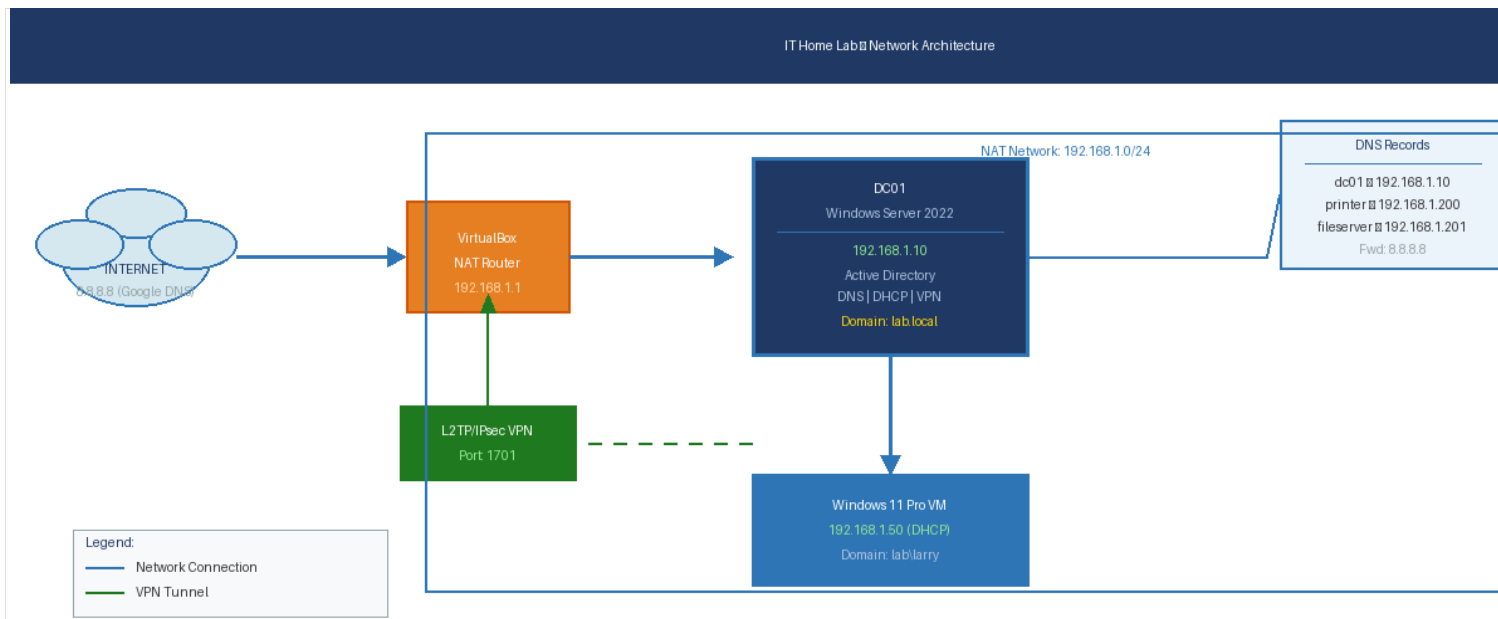


Figure 1.1: IT Home Lab Network Topology — showing all components and their relationships

1.3 Infrastructure Summary

Component	OS / Role	IP Address	Services Running
DC01 — Domain Controller	Windows Server 2022	192.168.1.10 (Static)	AD DS, DNS, DHCP, VPN (RRAS)
WIN11-CLIENT	Windows 11 Pro	192.168.1.50 (DHCP)	Domain member, VPN client
VirtualBox NAT	Virtual Router	192.168.1.1	Internet gateway, NAT
Google DNS (Forwarder)	External DNS	8.8.8.8 / 8.8.4.4	External DNS resolution

1.4 Technologies Demonstrated

Infrastructure	Security	Administration
Active Directory Domain Services	Windows Defender	Group Policy Management
DNS Server (Forward & Reverse)	Windows Firewall (Custom Rules)	User & Group Administration
DHCP Server (Scopes & Options)	IPsec / L2TP VPN Encryption	OU & Permission Management
VPN Server (RRAS / L2TP)	Security Event Log Analysis	Helpdesk Ticket Resolution
Azure for Students	UAC Configuration	DNS Record Management

**PHASE
1****Domain Controller & DNS Configuration**

2.1 Overview

The first phase established the foundational infrastructure of the lab. Windows Server 2022 was installed on DC01 with a static IP address of 192.168.1.10. Active Directory Domain Services was promoted, creating the lab.local domain. DNS was configured to support both internal name resolution and external internet access via forwarding.

2.2 DNS Manager Configuration

DNS Manager was accessed via Server Manager → Tools → DNS. DC01 was confirmed as the authoritative DNS server for the lab.local zone.

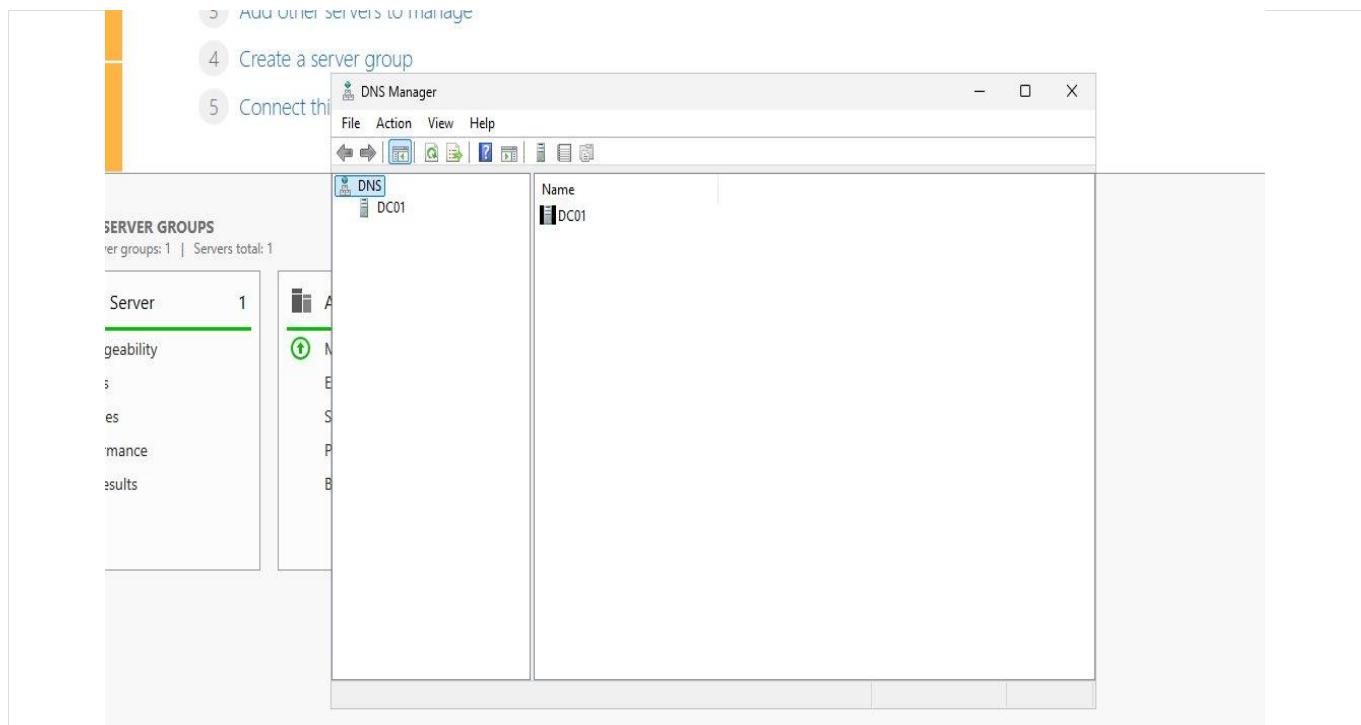


Figure 2.1: DNS Manager showing DC01 as the primary DNS server for lab.local

DNS Forwarder Configuration

A DNS forwarder was configured to redirect all external DNS queries (i.e. queries for domains not within lab.local) to Google's public DNS infrastructure. This enables internet access on all domain-joined machines without compromising internal domain resolution.

Configuration path: DNS Manager → Right-click DC01 → Properties → Forwarders tab → Edit → Add 8.8.8.8

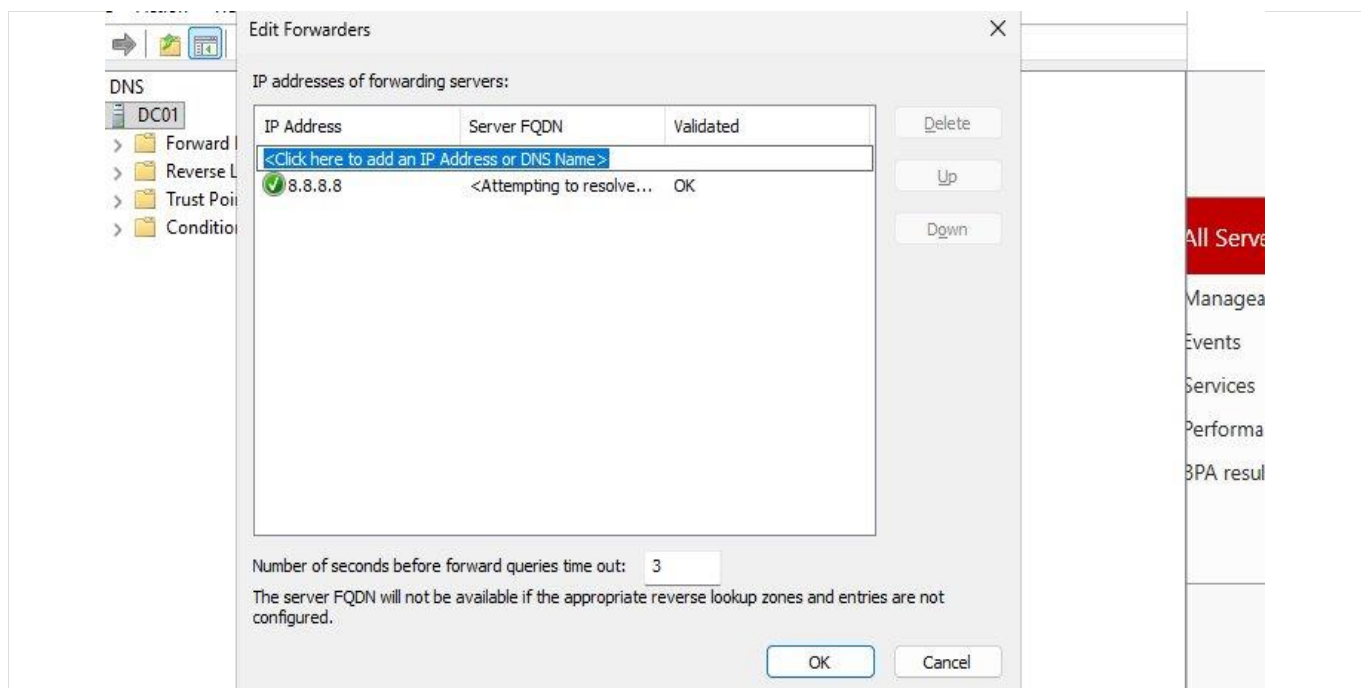


Figure 2.2: DNS Forwarder configured with 8.8.8.8 (Google DNS) — validated and active

Key Concept: Without DNS forwarders, domain-joined machines can resolve internal names (e.g. DC01.lab.local) but cannot resolve external internet addresses (e.g. google.com). The forwarder tells the DNS server to ask Google when it cannot answer the query itself — maintaining both internal domain functionality and internet access simultaneously.

2.3 Windows 11 Pro Installation

During initial setup, Windows 11 Home was identified on the client VM. Home edition does not support domain joining (System Properties shows the option greyed out with a restriction message). The VM was reinstalled with Windows 11 Pro using the same ISO — selecting "I don't have a product key" and choosing the Pro edition at the edition selection screen.

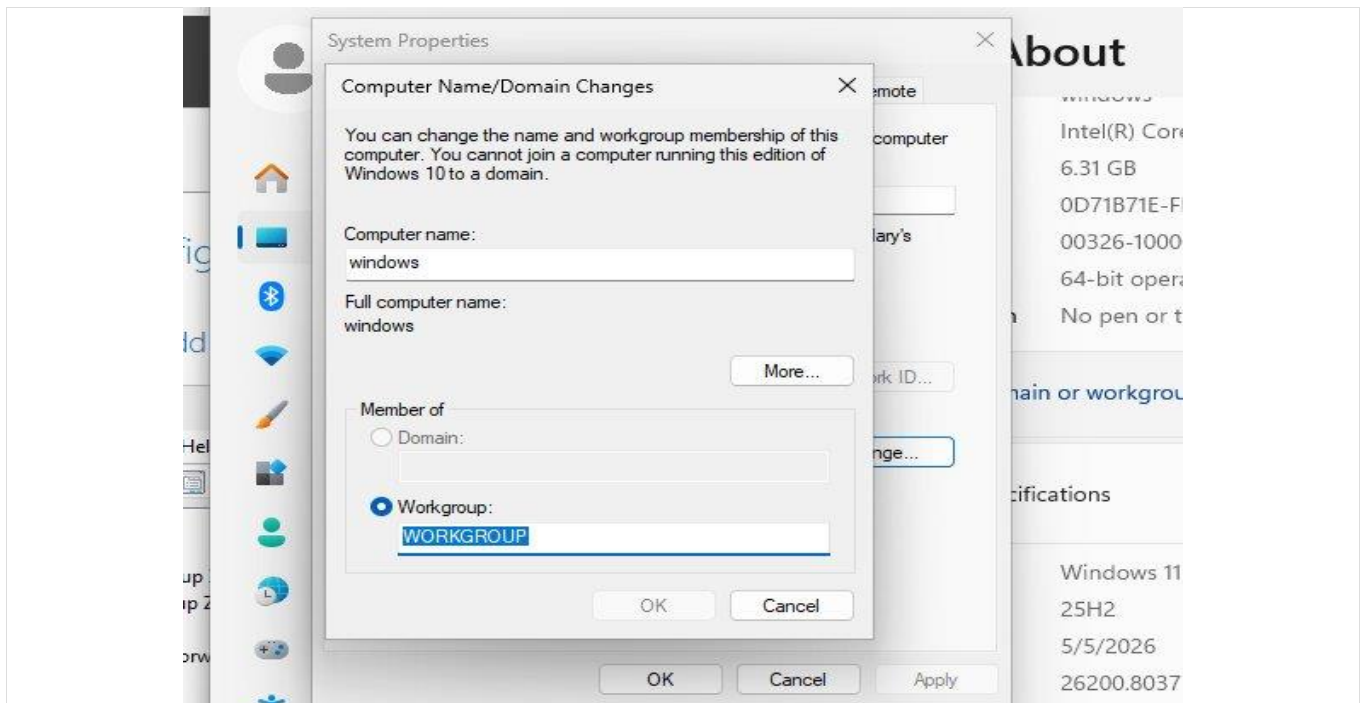


Figure 2.3: Windows 11 Home restriction — domain join unavailable, Pro edition required

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Active Directory Management

3.1 Overview

Active Directory Users and Computers (ADUC) was used to build out the organisational structure of the lab.local domain. This phase replicates the day-to-day AD administration tasks performed by IT Support staff, including user account management, security group assignment, and OU organisation.

3.2 Organisational Structure

The following structure was created within the lab.local → Company container:

Organisational Unit	Security Group	User Account	Purpose
IT	IT ADMIN	utsarga Dhakali (utsarga)	IT department users — elevated access
HR	HR ADMIN	Larry Wheels (larry)	HR department — restricted access

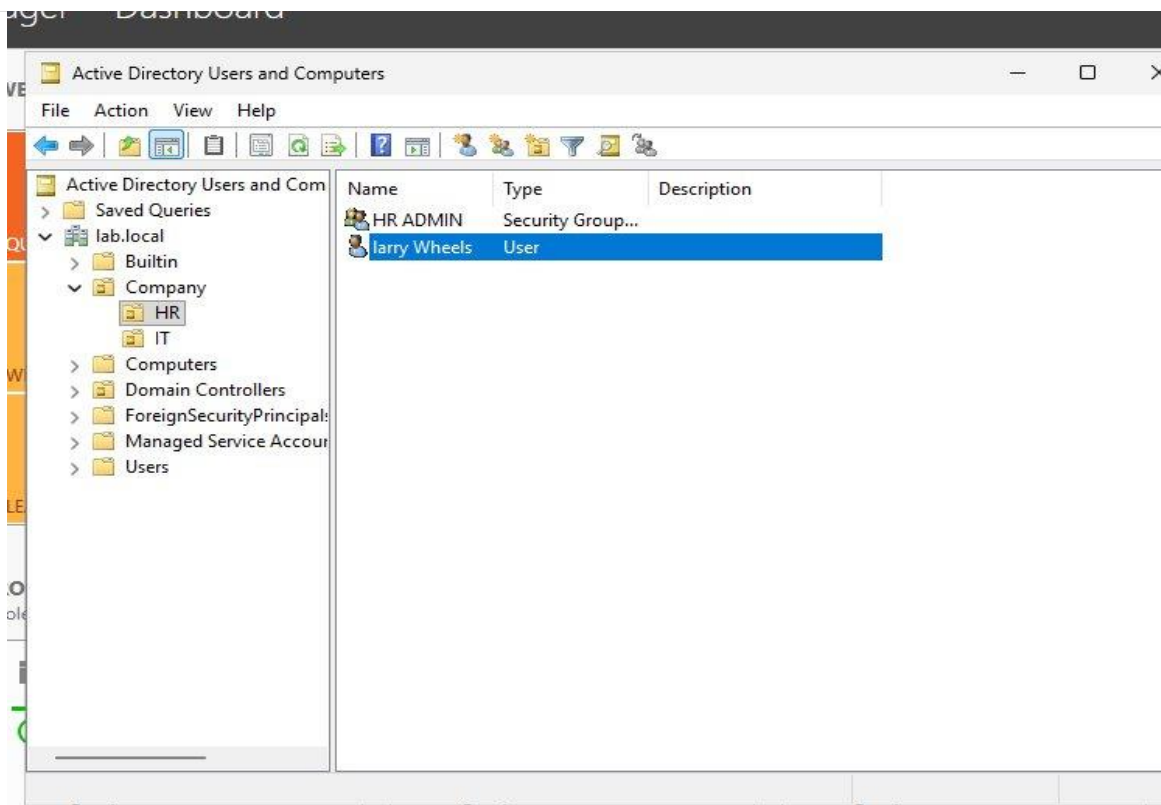


Figure 3.1: HR Organisational Unit containing HR ADMIN security group and Larry Wheels user account

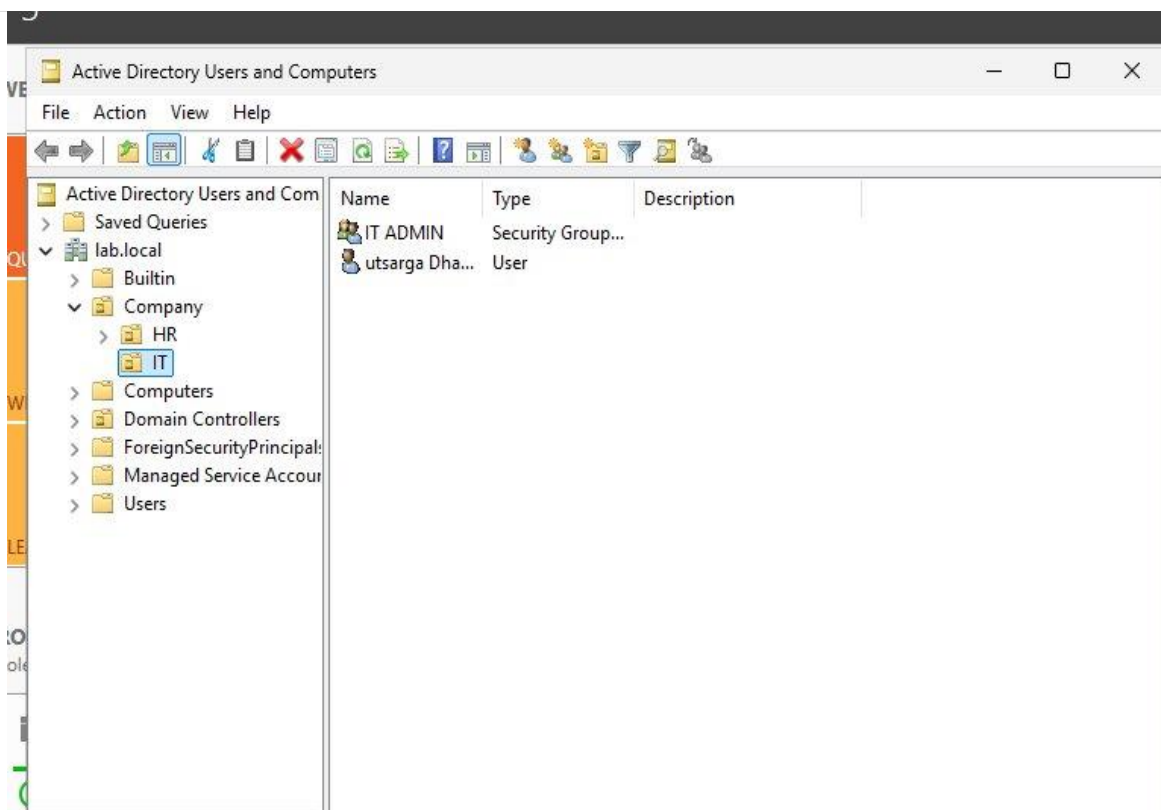


Figure 3.2: IT Organisational Unit containing IT ADMIN security group and utsarga Dhakali user account

3.3 Active Directory Administration Tasks

The following AD administration tasks were performed to simulate real-world IT support scenarios:

Task	Procedure
Create User Account	Right-click OU → New → User → Enter first name, last name, logon name → Set password → Configure account options
Reset User Password	Right-click user → Reset Password → Enter temporary password → Enable "User must change password at next logon"
Unlock Account	Right-click user → Properties → Account tab → Tick "Unlock account" → Apply
Disable / Enable Account	Right-click user → Disable Account (or Enable Account) — icon displays down arrow when disabled
Add User to Security Group	Double-click security group → Members tab → Add → Enter username → Check Names → OK
Move User Between OUs	Right-click user → Move → Select destination OU — GPO settings update automatically
Check Account Lockout	Right-click user → Properties → Account tab → "Unlock account" checkbox visible if locked
Set VPN Dial-in Permission	Right-click user → Properties → Dial-in tab → Network Access Permission → Allow access

Security Best Practice: When resetting a user password, always enable "User must change password at next logon." This ensures IT staff never know the user's actual password, maintaining the principle of least privilege and protecting user data privacy.

**PHASE
3****Group Policy Object (GPO) Management**

4.1 Overview

Group Policy Objects were created and linked to specific Organisational Units to enforce configuration settings across domain users and computers. GPOs are a fundamental tool for IT administrators to standardise desktop environments, enforce security policies, and restrict access to system functions.

4.2 Group Policy Management Console

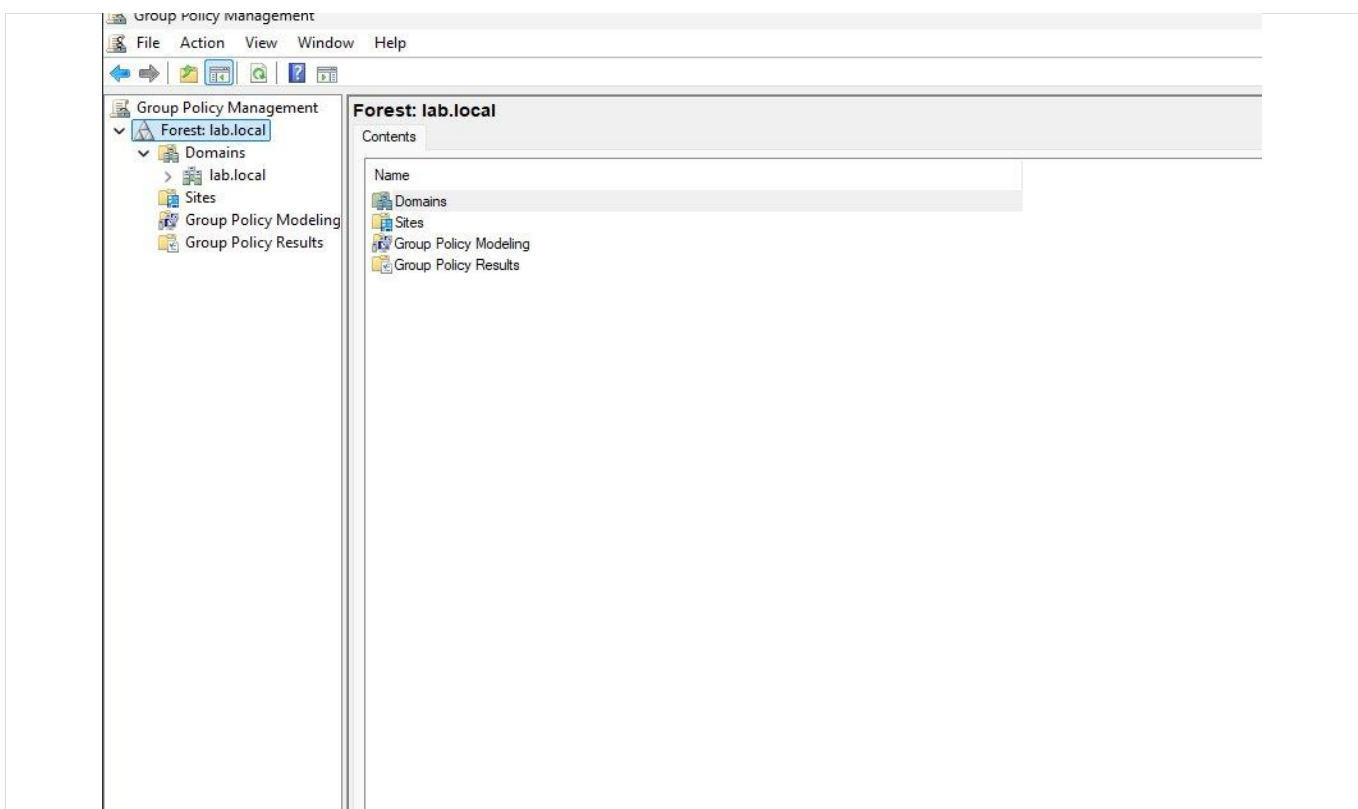


Figure 4.1: Group Policy Management Console showing the lab.local domain structure with linked GPOs

4.3 GPO 1 — IT Department Wallpaper Policy

A Group Policy Object was created and linked to the IT Organisational Unit to enforce a standard corporate desktop wallpaper. This prevents users from modifying the desktop background and ensures a consistent, professional appearance on all IT department workstations.

Configuration path: User Configuration → Policies → Administrative Templates → Desktop → Desktop → Desktop Wallpaper

Setting	Value
GPO Name	IT Wallpaper Policy
Linked To	IT Organisational Unit
Status	Enabled
Wallpaper Path	C:\Windows\Web\Wallpaper\Windows\img0.jpg
Wallpaper Style	Fill
Effect	Users cannot change desktop background — setting is greyed out

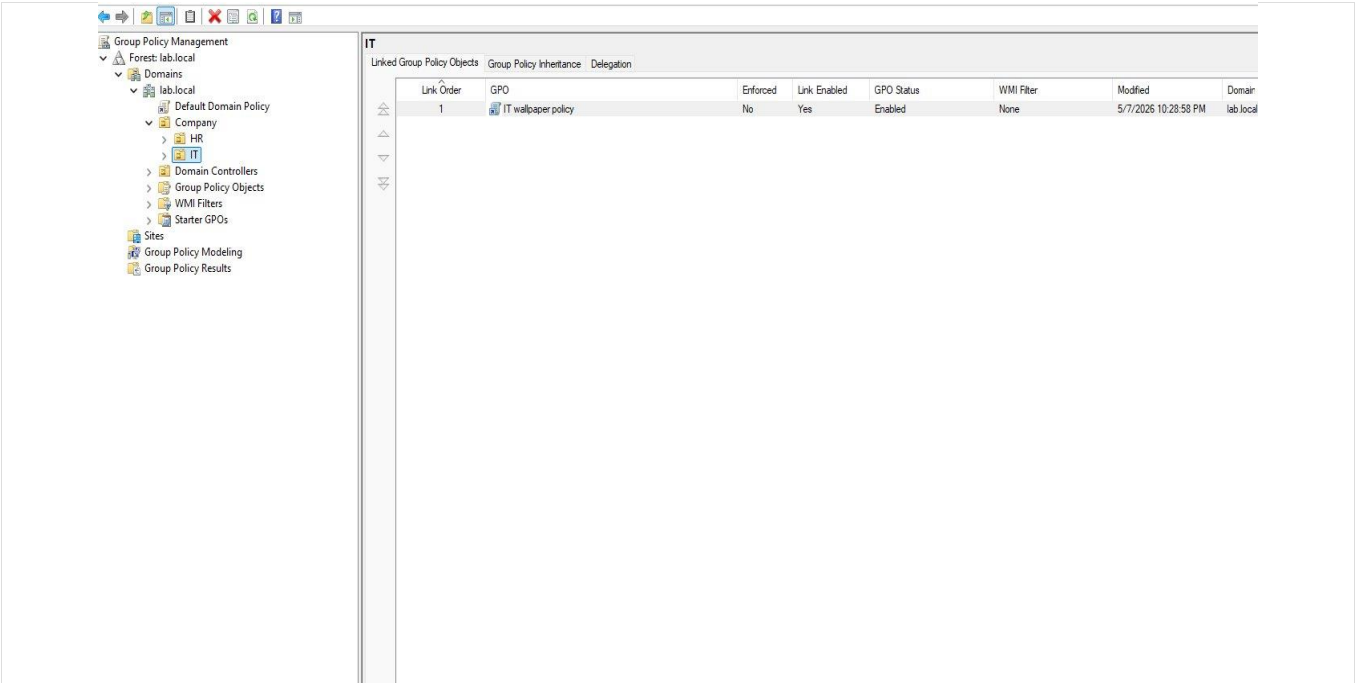


Figure 4.2: IT Wallpaper Policy linked to IT OU — Link Enabled: Yes, GPO Status: Enabled

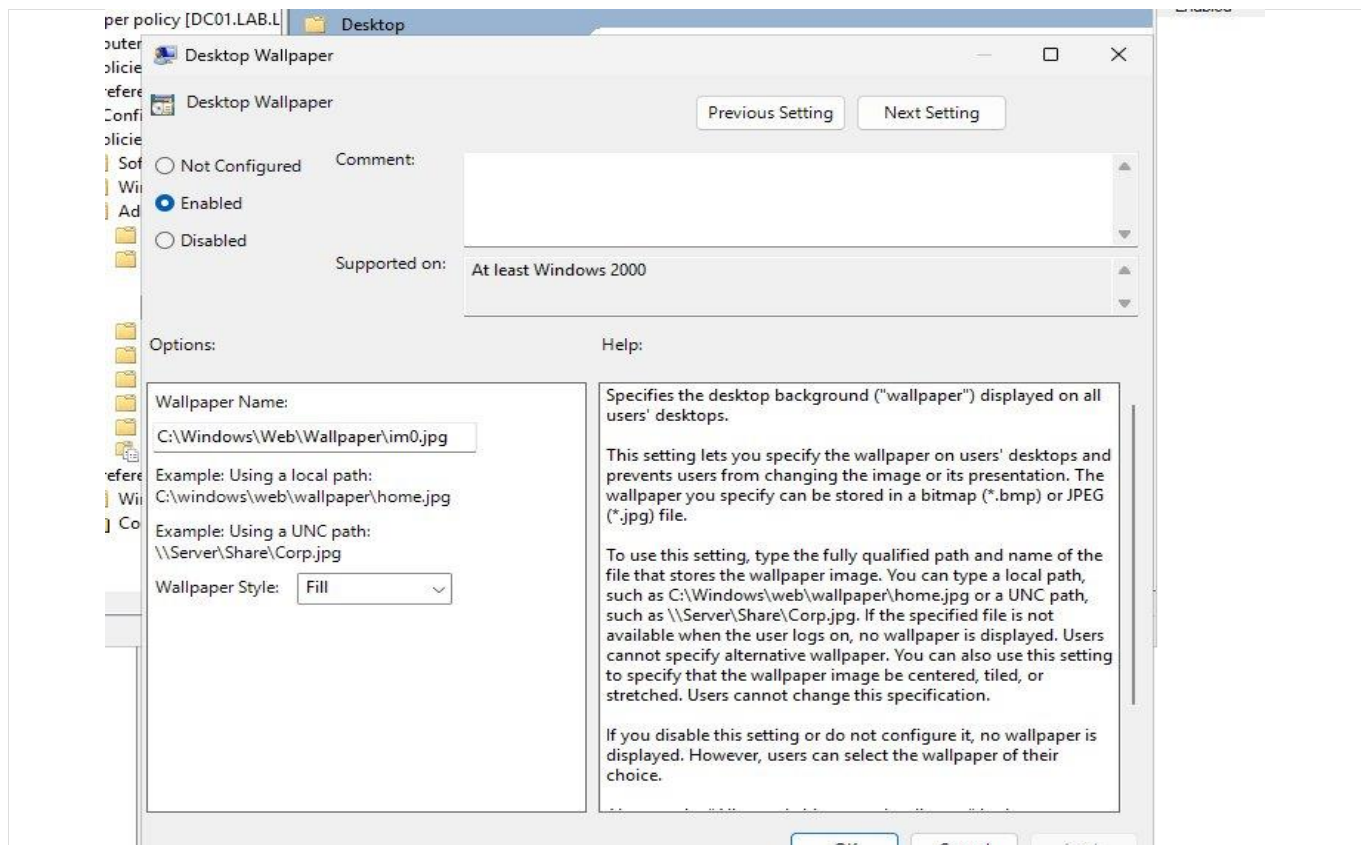


Figure 4.3: Desktop Wallpaper policy setting configured — Enabled with fill style

4.4 GPO 2 — HR Control Panel Restriction

A second GPO was created and linked to the HR Organisational Unit to prevent HR department users from accessing Control Panel and PC Settings. This is a standard enterprise security control that prevents non-technical staff from modifying system settings, installing software, or changing network configurations.

Configuration path: User Configuration → Policies → Administrative Templates → Control Panel → Prohibit access to Control Panel and PC settings

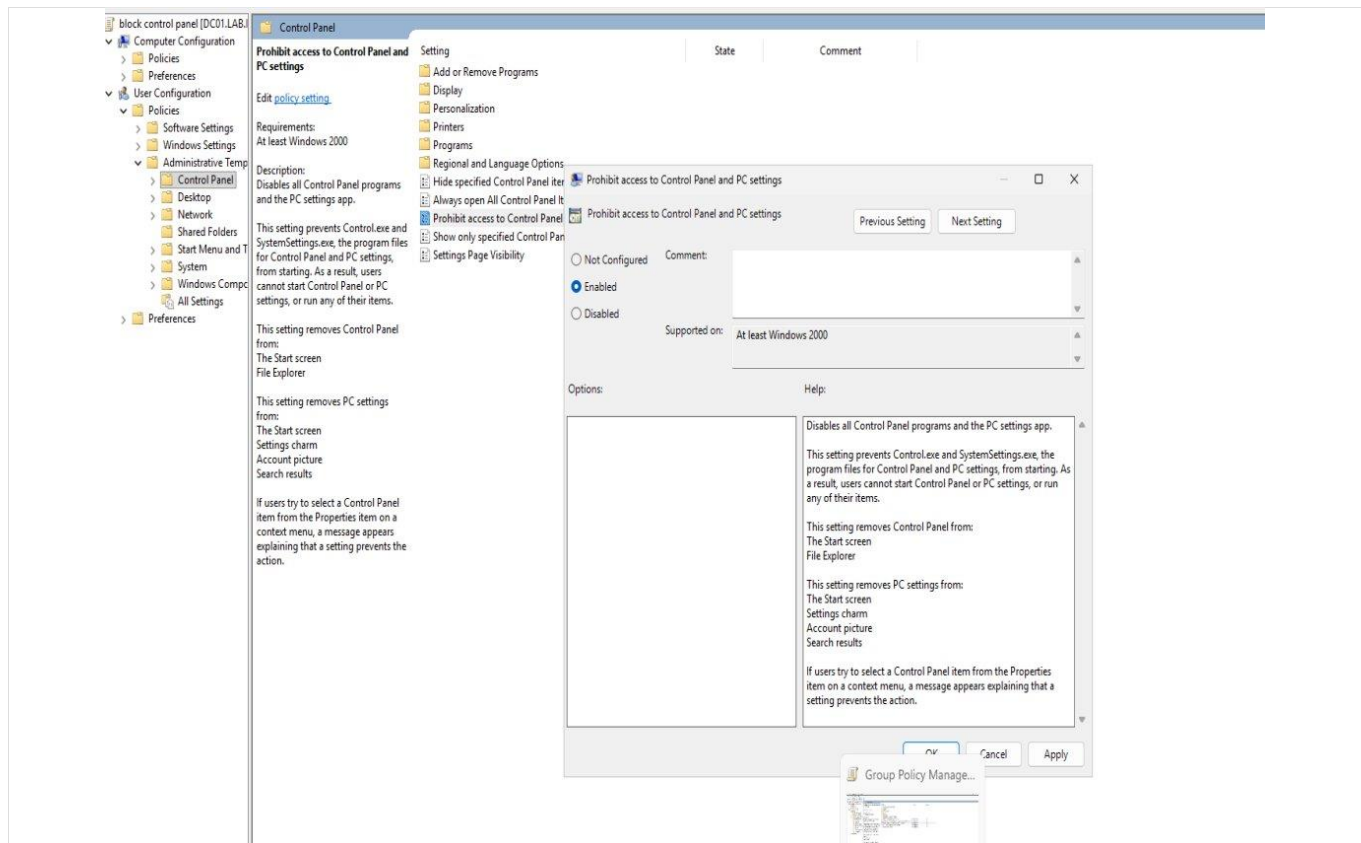


Figure 4.4: Block Control Panel GPO — Prohibit access to Control Panel and PC Settings enabled for HR OU

4.5 GPO Inheritance & Scope Verification

Group Policy inheritance was reviewed to confirm policies apply correctly. The Scope tab on each GPO displays all linked Organisational Units, Sites, and Security Groups. The Group Policy Inheritance tab on each OU shows all policies that apply, including those inherited from parent OUs.

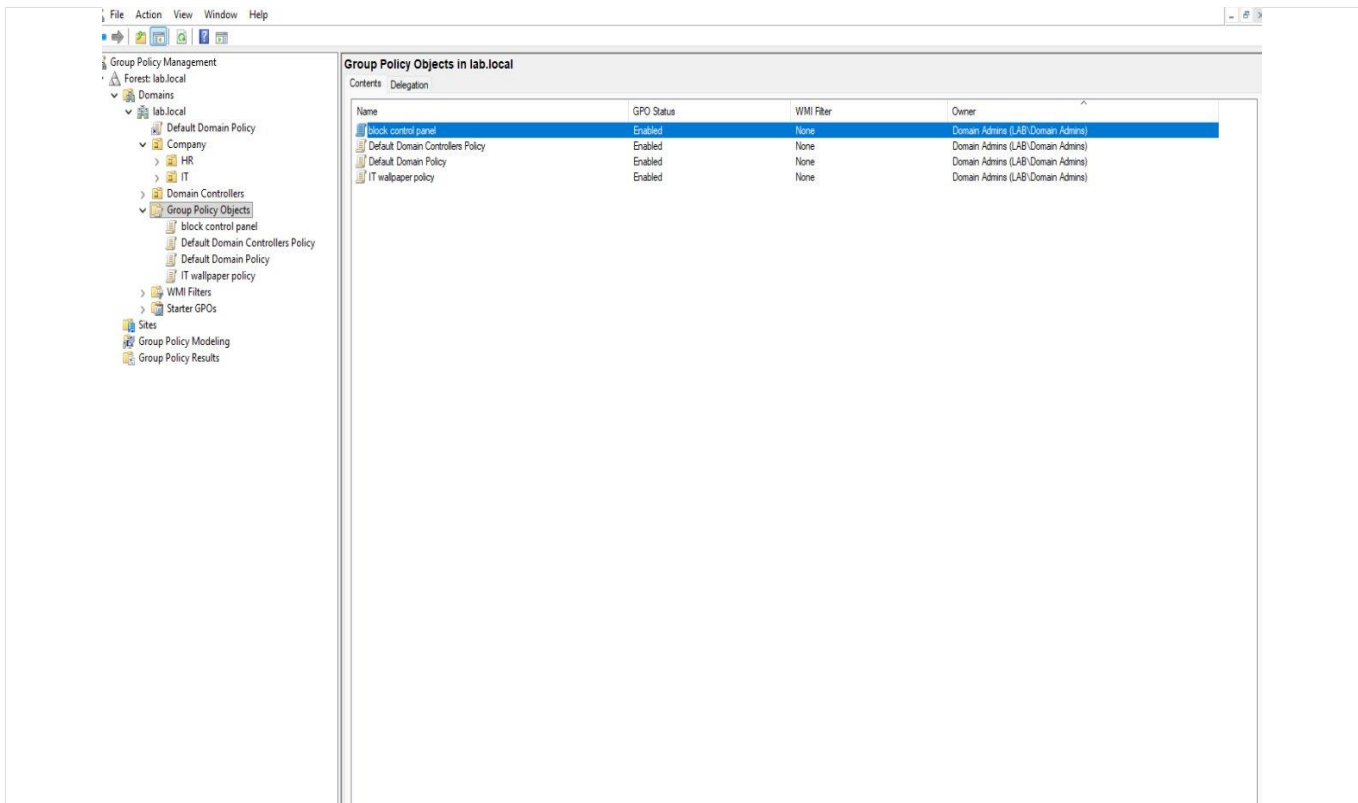


Figure 4.5: Group Policy Objects overview — all GPOs visible with status and domain

Key Command: Run `gpupdate /force` in an elevated Command Prompt to immediately apply Group Policy changes without waiting for the default 90-minute refresh interval. This is essential when testing GPO configurations during lab work or in a real support scenario.

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Networking — DHCP, DNS Records & Diagnostics

5.1 Overview

This phase covers the installation and configuration of DHCP on DC01, the creation of manual DNS records, and the practice of essential network diagnostic commands. These skills are used in daily IT support to troubleshoot connectivity issues, manage IP address assignment, and verify network service functionality.

5.2 DHCP Server Installation & Scope Configuration

The DHCP Server role was added to DC01 via Server Manager → Add Roles and Features. Following installation, the service was authorised in Active Directory and a DHCP scope was created to automatically assign network configuration to client machines.

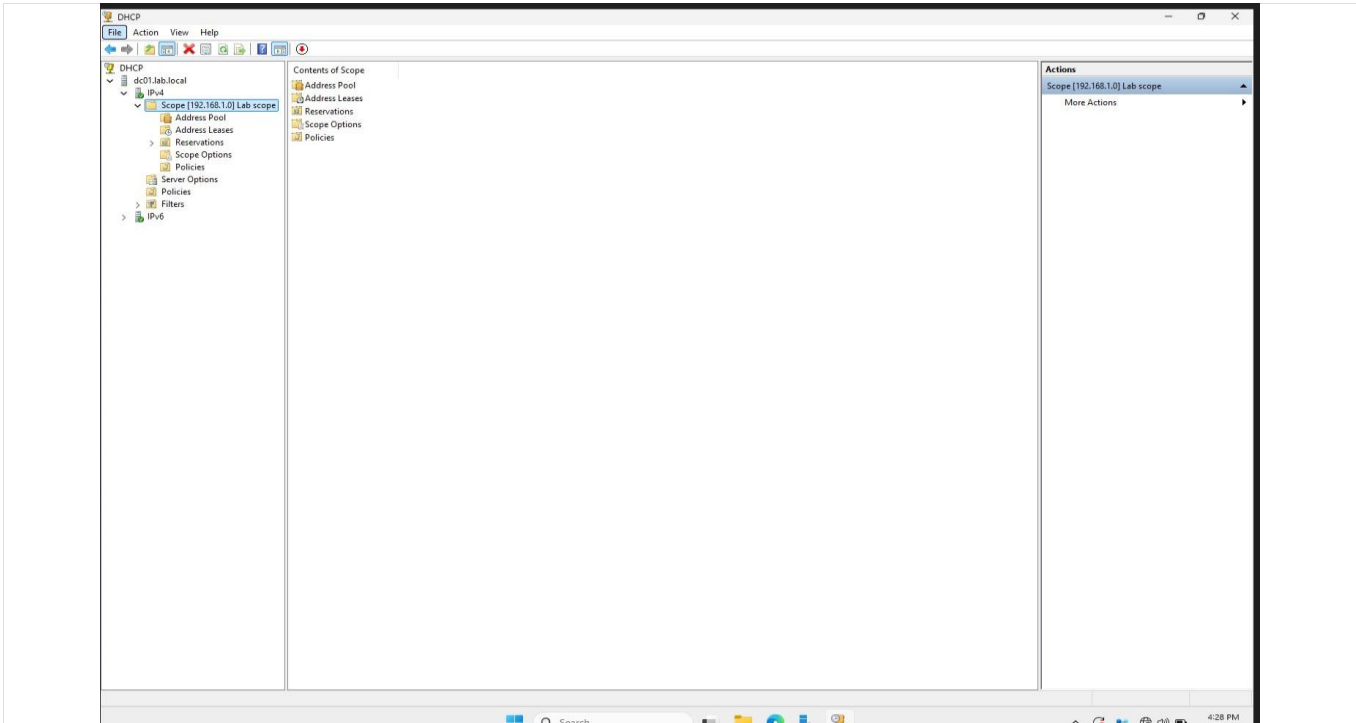


Figure 5.1: DHCP Manager showing Lab Scope created and active under IPv4

DHCP Scope Configuration Details

DHCP Option	Value	Purpose
-------------	-------	---------

Scope Name	Lab Scope	Identifies this DHCP scope in the management console
IP Range Start	192.168.1.50	First address available for dynamic assignment
IP Range End	192.168.1.100	Last address available for dynamic assignment
Subnet Mask	255.255.255.0	Defines the network boundary (/24)
003 — Default Gateway	192.168.1.1	VirtualBox NAT router — provides internet routing
006 — DNS Server	192.168.1.10	Points clients to DC01 for domain name resolution
015 — DNS Domain Name	lab.local	Appended to unqualified hostnames for DNS searches
Lease Duration	8 days	Period before client must renew the IP lease

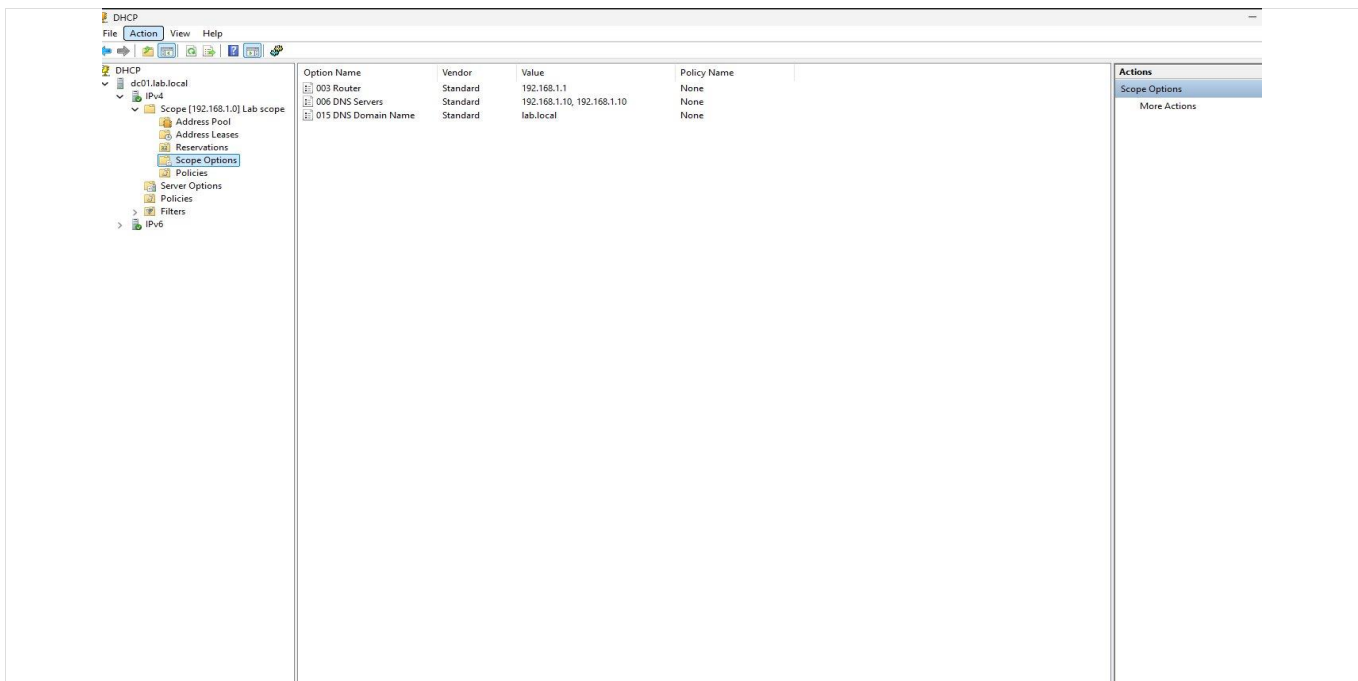


Figure 5.2: Scope Options — Router (003), DNS Server (006) and Domain Name (015) all configured

Real-World Note: In production environments, static IP addresses are reserved for servers, printers, and network infrastructure. Client workstations use DHCP. DHCP reservations (binding a MAC address to a specific IP) can be used for devices that need a consistent address — such as printers — without manually configuring a static IP on the device itself.

5.3 Network Diagnostic Commands

The following commands were executed on the Windows 11 client VM to validate network configuration and practice diagnostic techniques used in IT support:

```

C:\WINDOWS\system32> ipconfig /renew

C:\WINDOWS\system32> ipconfig /all

C:\WINDOWS\system32> ipconfig /all

```

The output of the `ipconfig /all` command confirms the network configuration. Key details include:

- IP Address:** 192.168.1.4 (Preferred)
- DNS Suffix:** lab.local
- DNS Servers:** 192.168.1.1
- Default Gateway:** 192.168.1.1
- Physical Address (MAC):** 08-00-27-35-F8-DF
- DHCP Enabled:** Yes
- DNS Proxy Enabled:** No
- DNS Suffix Search List:** lab.local
- Adapter:** Intel(R) PRO/1000 MT Desktop Adapter
- IPv6 Address:** fe80::dadd:a6:fe06:11d3%11 (Preferred)
- IPv6 Address:** 192.168.1.4 (Preferred)
- IPv6 Subnet Mask:** 255.255.255.0
- IPv6 Address Obtained:** Tuesday, May 12, 2026 9:43:15 AM
- IPv6 Address Expires:** Tuesday, May 12, 2026 9:53:15 AM
- IPv6 Default Gateway:** 192.168.1.1
- IPv6 Server:** 192.168.1.2
- IPv6 IAID:** 101187623
- IPv6 Client DUID:** 00-01-01-00-31-90-15-E9-08-00-27-35-F8-DF
- DNS Servers:** 192.168.1.1
- NetBIOS over Tcpip:** Enabled

Figure 5.3: `ipconfig /all` output — confirms DHCP-assigned IP (192.168.1.50), DNS pointing to DC01 (192.168.1.10), and domain suffix lab.local

TCP	0.0.0.0:49669	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49670	0.0.0.0:0	LISTENING
TCP	192.168.1.50:139	0.0.0.0:0	LISTENING
TCP	192.168.1.50:49678	192.168.1.10:445	ESTABLISHED
TCP	192.168.1.50:51130	150.171.69.254:443	ESTABLISHED
TCP	192.168.1.50:51478	14.200.100.122:80	TIME_WAIT
TCP	192.168.1.50:51479	14.200.100.122:80	TIME_WAIT
TCP	192.168.1.50:51480	14.200.100.122:80	TIME_WAIT
TCP	192.168.1.50:51489	72.145.35.119:443	TIME_WAIT
TCP	192.168.1.50:51501	172.172.255.218:443	ESTABLISHED
TCP	192.168.1.50:51504	20.190.167.20:443	TIME_WAIT
TCP	192.168.1.50:52817	13.107.42.10:443	ESTABLISHED
TCP	192.168.1.50:54583	52.123.128.254:443	ESTABLISHED
TCP	192.168.1.50:55580	150.171.28.11:443	ESTABLISHED
TCP	192.168.1.50:55581	204.79.197.203:80	TIME_WAIT
TCP	192.168.1.50:55582	199.232.214.172:80	TIME_WAIT
TCP	192.168.1.50:55583	202.7.177.249:443	ESTABLISHED
TCP	192.168.1.50:55595	20.190.167.20:443	ESTABLISHED
TCP	192.168.1.50:55596	20.190.167.20:443	ESTABLISHED
TCP	192.168.1.50:55597	20.190.167.20:443	ESTABLISHED
TCP	192.168.1.50:55598	20.190.167.20:443	ESTABLISHED
TCP	192.168.1.50:56406	52.110.12.41:443	ESTABLISHED
TCP	192.168.1.50:56407	52.110.12.1:443	ESTABLISHED
TCP	192.168.1.50:56413	150.171.109.17:443	TIME_WAIT
TCP	192.168.1.50:56423	150.171.109.17:443	TIME_WAIT
TCP	192.168.1.50:56424	4.230.171.124:443	TIME_WAIT

Figure 5.4: netstat -an output — shows ESTABLISHED connections to Microsoft servers from 192.168.1.50, confirming network connectivity

Command	Function	Common Use Case
ipconfig /all	Displays full network adapter configuration	Verify IP, DNS, gateway, MAC address, DHCP status
ipconfig /release	Releases DHCP-assigned IP address	Force return of IP lease before requesting new one
ipconfig /renew	Requests new IP from DHCP server	Refresh IP configuration after network changes
ipconfig /flushdns	Clears local DNS resolver cache	Fix stale DNS entries causing wrong resolution
ping <IP or hostname>	Tests connectivity to a host	Ping IP first, then hostname to isolate DNS vs network
nslookup <name>	Queries DNS for name resolution	Verify DNS records resolve correctly
tracert <host>	Traces route packets take to destination	Identify where network connectivity breaks
netstat -an	Lists all active network connections	Identify active connections, listening ports, suspicious traffic

5.4 DNS A Record Creation

Manual DNS A records were created in the lab.local Forward Lookup Zone to simulate the real-world process of registering network devices such as printers and file servers in DNS. This allows users to access devices by name rather than memorising IP addresses.

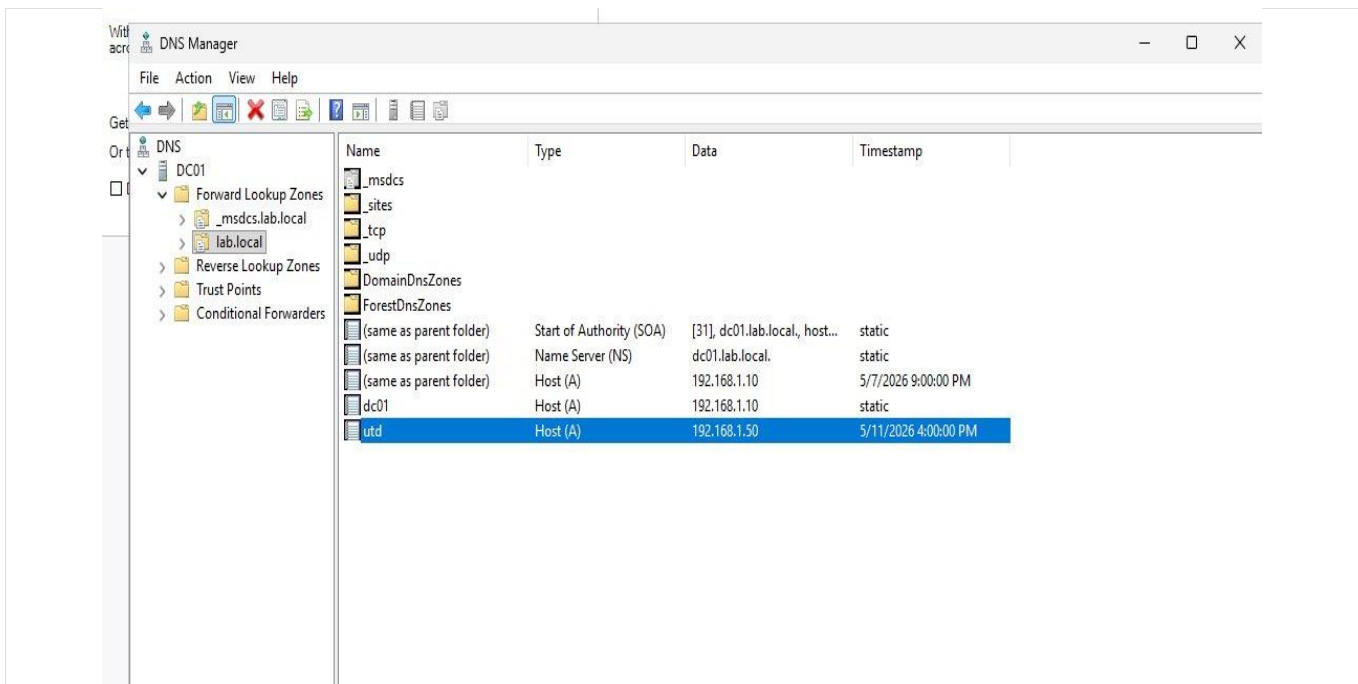


Figure 5.5: lab.local DNS zone showing A records — dc01 (192.168.1.10), utd/client (192.168.1.50), printer (192.168.1.200), fileserver (192.168.1.201)

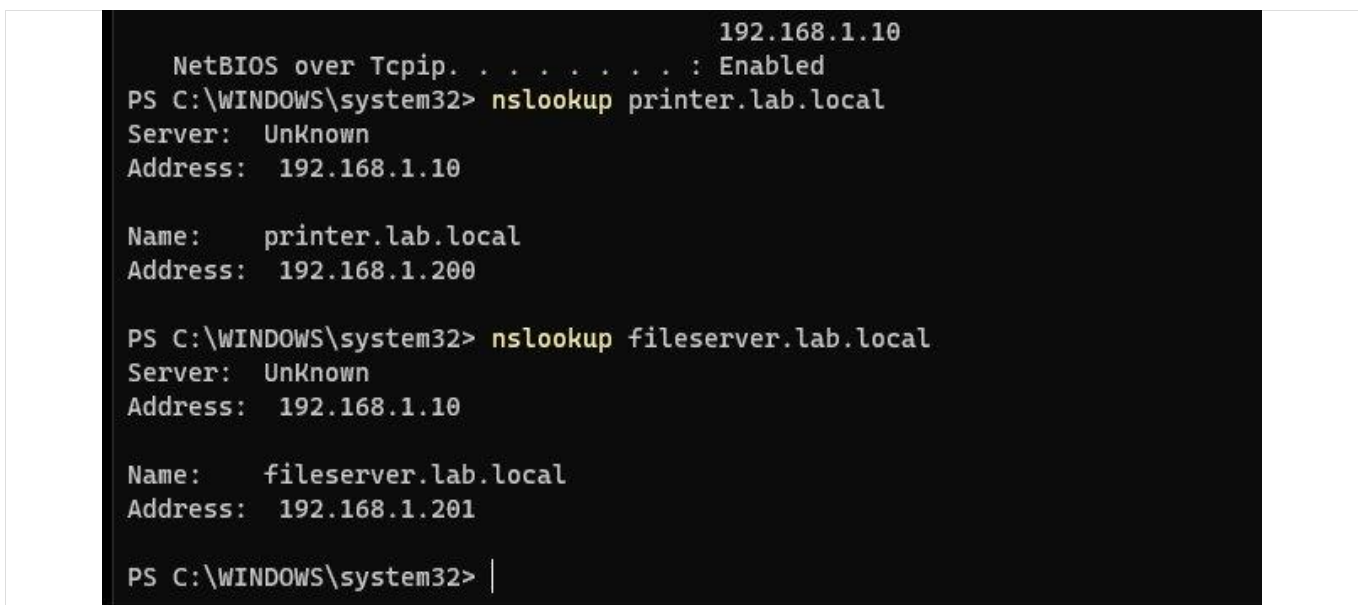


Figure 5.6: nslookup confirming printer.lab.local resolves to 192.168.1.200 and fileserver.lab.local to 192.168.1.201

**PHASE
5****Helpdesk Ticket Simulation**

6.1 Overview

Five real-world IT support scenarios were simulated to practise the structured troubleshooting and communication skills required for Level 1 and Level 2 helpdesk roles. Each ticket was worked through from initial user report through to documented resolution, following the standard ITIL-aligned support process.

Standard Helpdesk Process: (1) Verify user identity → (2) Gather information and reproduce the issue → (3) Troubleshoot systematically from simplest to most complex → (4) Resolve and confirm with user → (5) Document the ticket with cause and resolution.

Ticket #01 Password Reset Request

User Report:

"Hi IT, this is Sarah from HR. I cannot log into my computer this morning. It says my password is incorrect but I am certain I am typing it correctly. I have not changed anything. Can you help?"

Troubleshooting Steps:

1. Verified user identity — requested full name and employee ID before making any changes
2. Opened Active Directory Users and Computers (ADUC) → navigated to HR OU → located Sarah Smith
3. Opened user Properties → Account tab — checked for account lockout ("Unlock account" checkbox) and password expiry
4. Right-clicked user → Reset Password → set temporary password → enabled "User must change password at next login"
5. Communicated temporary password to user via secure channel (not email in plaintext)
6. Followed up to confirm successful login and password change

Resolution: Password reset in Active Directory. User prompted to set new personal password on next login. Ticket closed after user confirmed successful access.

Ticket #02 Cannot Access Shared Drive

User Report:

"Hi IT, this is John from IT. My internet is working fine but I cannot access the shared drive. It was working yesterday and I have not changed anything."

Troubleshooting Steps:

7. Asked clarifying questions: when did it start, any error message displayed, is anyone else affected
8. Ran ipconfig /all on John's machine — confirmed correct IP, DNS pointing to 192.168.1.10
9. Ran ping fileserver.lab.local — succeeded, confirming DNS resolution is working
10. Ran ping 192.168.1.201 — succeeded, confirming basic network connectivity to server
11. Checked John's user account in ADUC → Member Of tab — confirmed he had been removed from the file server access security group
12. Added John back to the correct security group → confirmed access restored

Resolution: User was inadvertently removed from the security group controlling shared drive access. Re-added to group in ADUC. Shared drive access immediately restored. Ticket closed.

Ticket #03 New Employee Cannot Log In**User Report:**

"Hi IT, I am Emma Johnson and I am starting today. My manager said my account should be ready but I cannot log in at all. The screen just says my username or password is incorrect."

Troubleshooting Steps:

13. Searched ADUC for "Emma Johnson" — confirmed account existed in the correct OU
14. Checked Account tab → confirmed account was disabled (overlooked during onboarding)
15. Enabled the account → checked it was not locked and had not expired
16. Verified Emma was a member of the correct security groups for her department
17. Reset password and enabled "User must change password at next logon"
18. Communicated username (ejohnson@lab.local) and temporary password to Emma and her manager

Resolution: New user account was disabled. Account enabled in ADUC, placed in correct OU and security groups, password set. Emma successfully logged in. Onboarding process flagged for improvement to prevent recurrence.

Ticket #04 No Internet Access — DNS Issue**User Report:**

"Hi IT, my laptop is connected to the company WiFi but I cannot open any websites at all. My colleague sitting next to me has full internet access. Everything was working this morning."

Troubleshooting Steps:

19. Clarified: how long, any error message, any recent changes to the device
20. Ran ping 8.8.8.8 — succeeded, confirming physical network connectivity and routing are intact
21. This immediately isolated the issue as a DNS problem, not a connectivity problem

22. Ran `ipconfig /all` — DNS server field showed incorrect address (not 192.168.1.10)
23. Ran `ipconfig /release` then `ipconfig /renew` — DHCP reassigned correct DNS server
24. Ran `ipconfig /flushdns` to clear any stale cached DNS entries
25. Ran `nslookup google.com` — confirmed correct resolution, internet access restored

Resolution: DHCP lease had assigned incorrect DNS configuration. DHCP renewal corrected the settings. Internet access restored. Underlying DHCP issue investigated and resolved on DC01.

Ticket #05 Account Locked After Extended Leave

User Report:

"Hi IT, I just came back from three weeks holiday and my computer will not let me log in. The screen says my account has been locked."

Troubleshooting Steps:

26. Verified user identity before touching the account
27. Opened ADUC → located user → Account tab → ticked "Unlock account" → Applied
28. Opened Event Viewer on DC01 → Windows Logs → Security → filtered for Event ID 4740 (account lockout)
29. Event log identified the lockout source: user's mobile phone attempting to sync email with old cached credentials
30. Confirmed account was also approaching password expiry — proactively reset password
31. Advised user to update saved credentials on all devices (phone, tablet) to prevent recurrence

Resolution: Account unlocked in ADUC. Password proactively reset. Root cause identified as mobile device using cached credentials post-holiday. User educated on updating credentials across all devices. Ticket closed.

**PHASE
7****VPN Setup — L2TP/IPsec Remote Access**

7.1 Overview

A VPN server was configured on DC01 using the Routing and Remote Access Service (RRAS) role. This enables domain users to securely connect to the lab.local network from outside the office environment, replicating the remote access infrastructure used in real-world enterprise environments.

The L2TP/IPsec protocol was selected for its balance of compatibility and security. L2TP (Layer 2 Tunneling Protocol) creates the tunnel, while IPsec (Internet Protocol Security) provides encryption and authentication using a pre-shared key.

7.2 VPN Server Configuration — DC01

The following steps were performed on DC01 to establish the VPN server:

#	Step	Detail
1	Install RRAS Role	Server Manager → Add Roles and Features → Remote Access → DirectAccess and VPN (RAS)
2	Launch RRAS Console	Server Manager → Tools → Routing and Remote Access
3	Configure VPN	Right-click DC01 → Configure and Enable Routing and Remote Access → Deploy VPN only → Custom Configuration → VPN Access
4	Enable L2TP Ports	Ports → Properties → WAN Miniport (L2TP) → Configure → Enable Remote access connections (inbound only) → 5 ports
5	Set Pre-shared Key	Right-click DC01 → Properties → Security tab → Allow custom IPsec policy for L2TP/IKEv2 → Enter pre-shared key: VPNlab123
6	Grant User Access	ADUC → larry Wheels → Properties → Dial-in tab → Network Access Permission → Allow access
7	Restart Service	Right-click DC01 (local) → All Tasks → Restart

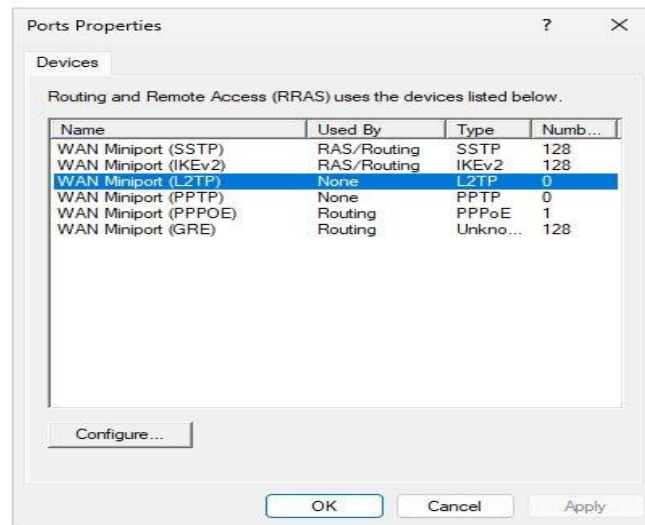


Figure 7.1: Ports Properties showing L2TP configured with 5 ports for RAS — Used By: RAS, Number: 5

7.3 VPN Client Configuration — Windows 11 VM

The Windows built-in VPN client was configured on the Windows 11 VM to connect to the VPN server on DC01:

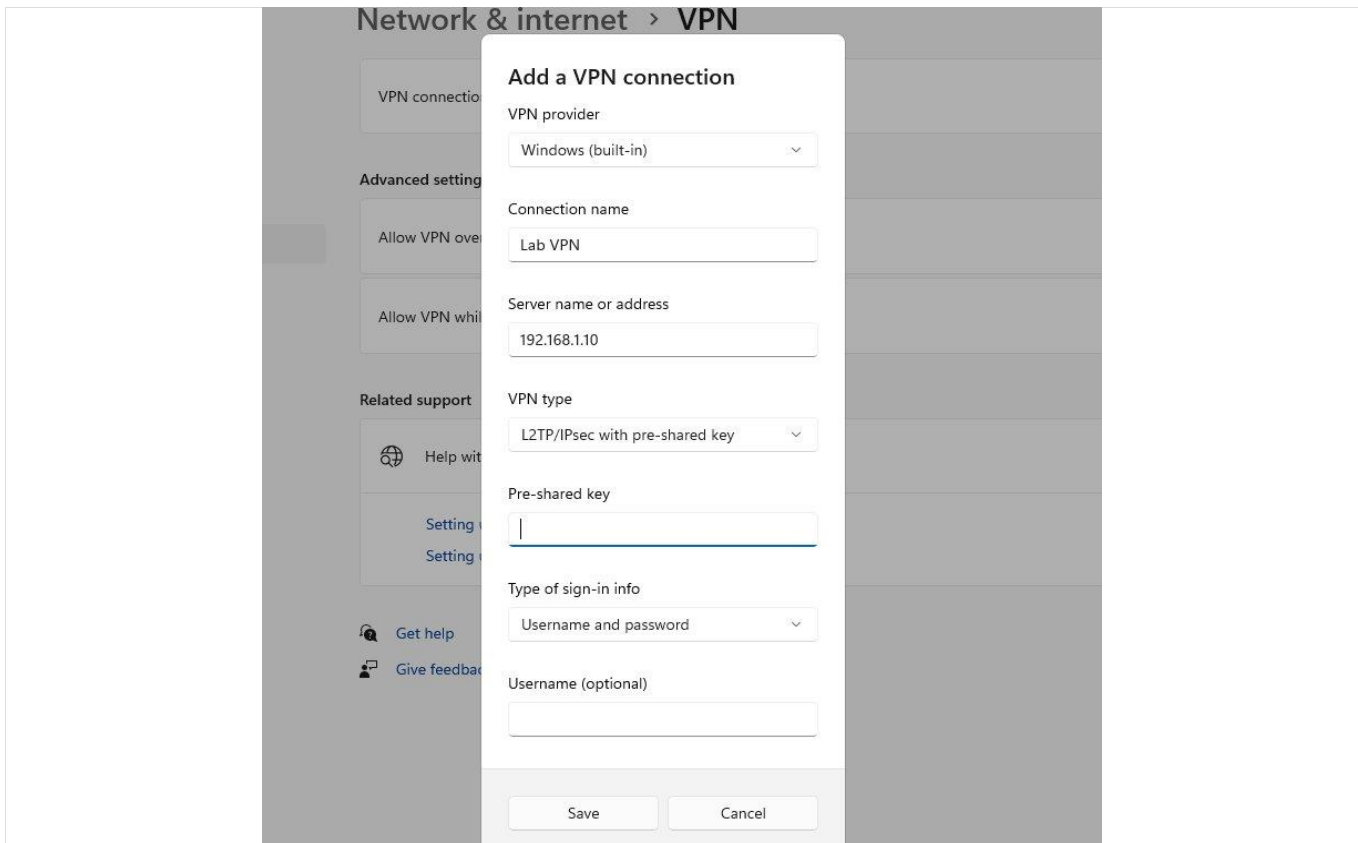


Figure 7.2: VPN connection configured — L2TP/IPsec with pre-shared key, credentials stored

Registry Fix — L2TP Behind NAT

Windows 11 blocks L2TP VPN connections when operating behind a NAT device (such as the VirtualBox NAT network) by default. The following registry modification was required to enable the connection:

```
reg add "HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\PolicyAgent" /v AssumeUDPEncapsulationContextOnSendRule /t REG_DWORD /d 2 /f
```

The value /d 2 instructs Windows to allow L2TP encapsulation even when the client is behind a NAT device. A full system restart was required for the change to take effect. The registry value was verified with:

```
reg query "HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\PolicyAgent" /v AssumeUDPEncapsulationContextOnSendRule
```

7.4 Successful VPN Connection

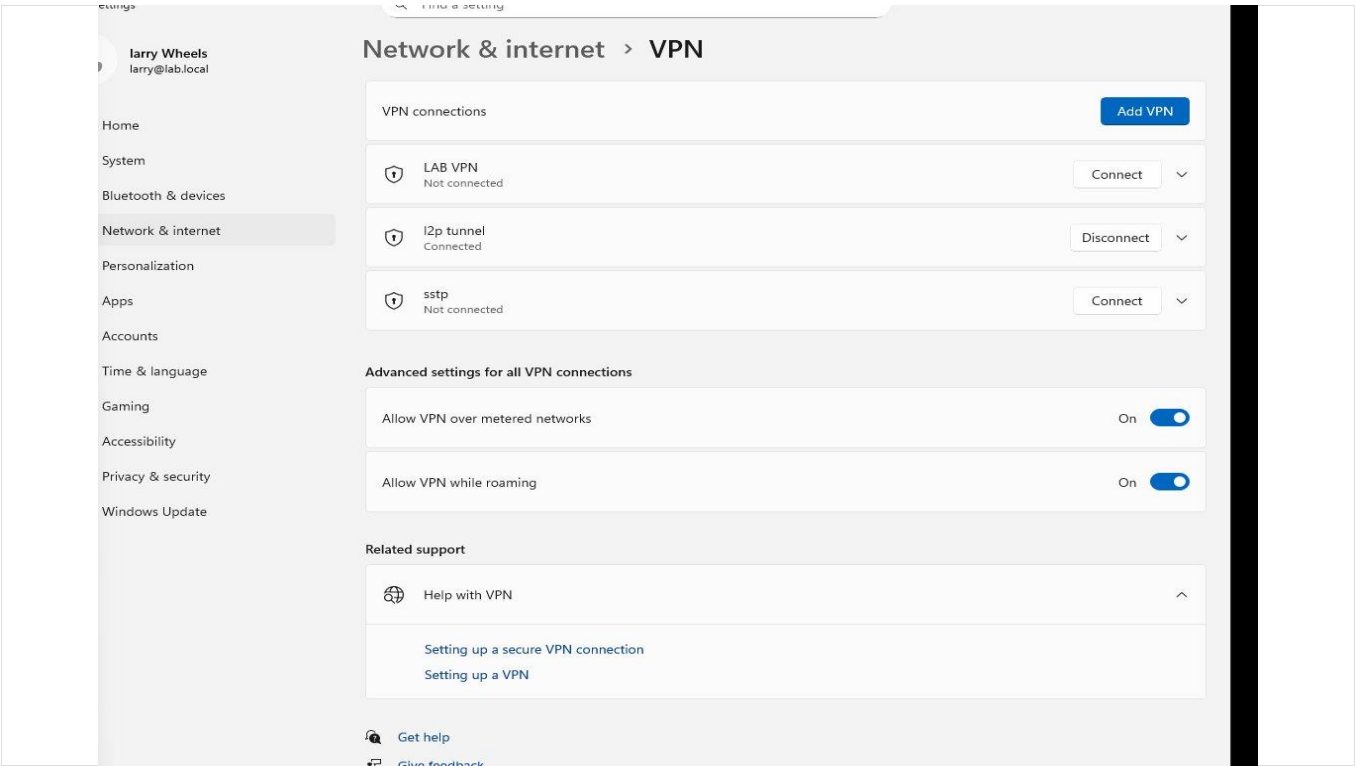


Figure 7.3: L2TP tunnel successfully connected — larry@lab.local authenticated and active

7.5 Troubleshooting Log

The VPN configuration required several troubleshooting cycles, providing valuable real-world diagnostic experience:

Issue Encountered	Root Cause	Resolution
L2TP security layer error	Registry not applied / VM not restarted	Applied registry fix, restarted VM
IKE authentication credentials unacceptable	IKEv2 requires certificates, not pre-shared key	Reverted to L2TP/IPsec protocol
No L2TP ports available	L2TP WAN Miniport had 0 ports configured	Enabled Remote access connections on L2TP miniport, set 5 ports
User Dial-in access denied	Larry's Dial-in permission was set to Deny	Changed to Allow access in ADUC → Dial-in tab
Connection succeeded	All issues resolved systematically	L2TP tunnel active, authenticated via AD

Outcome: A fully functional L2TP/IPsec VPN server was successfully configured on DC01. Domain users with Dial-in permission can authenticate through Active Directory and access lab.local resources remotely. This directly demonstrates the job requirement: "Configure and maintain VPN connections to support secure remote access for users."

PHASE
10

Security Fundamentals

8.1 Overview

This phase covers the core security tools and concepts used in enterprise IT support environments. Effective security management requires both the technical ability to configure protection tools and the knowledge to recognise, respond to, and document security incidents.

8.2 Security Event Log — Event ID Reference

Windows Security Event Logs were reviewed in Event Viewer (Server Manager → Tools → Event Viewer → Windows Logs → Security) on DC01. The VPN authentication events from the successful L2TP connection in Phase 7 were identifiable as Event ID 4624 entries.

Event ID	Meaning	IT Support Action Required
4624	Successful account login	Normal — monitor for logins at unusual hours or from unexpected locations
4625	Failed login attempt	Investigate if repeated — may indicate brute force attack or forgotten password
4740	User account locked out	Unlock account in ADUC → investigate source computer → check for cached credentials
4720	New user account created	Verify was authorised — alert if unexpected during off-hours
4726	User account deleted	Verify authorised offboarding — investigate immediately if unexpected
4728	User added to security group	Verify authorised — privilege escalation risk if unexpected
4648	Logon with explicit credentials	May indicate credential reuse — investigate if repeated or unexpected
4634	Account logoff	Normal — useful for audit trails and session duration tracking

8.3 Windows Defender

Windows Security was reviewed on the Windows 11 client VM. A Quick Scan was performed confirming zero threats. Security intelligence (virus definitions) was confirmed as current.

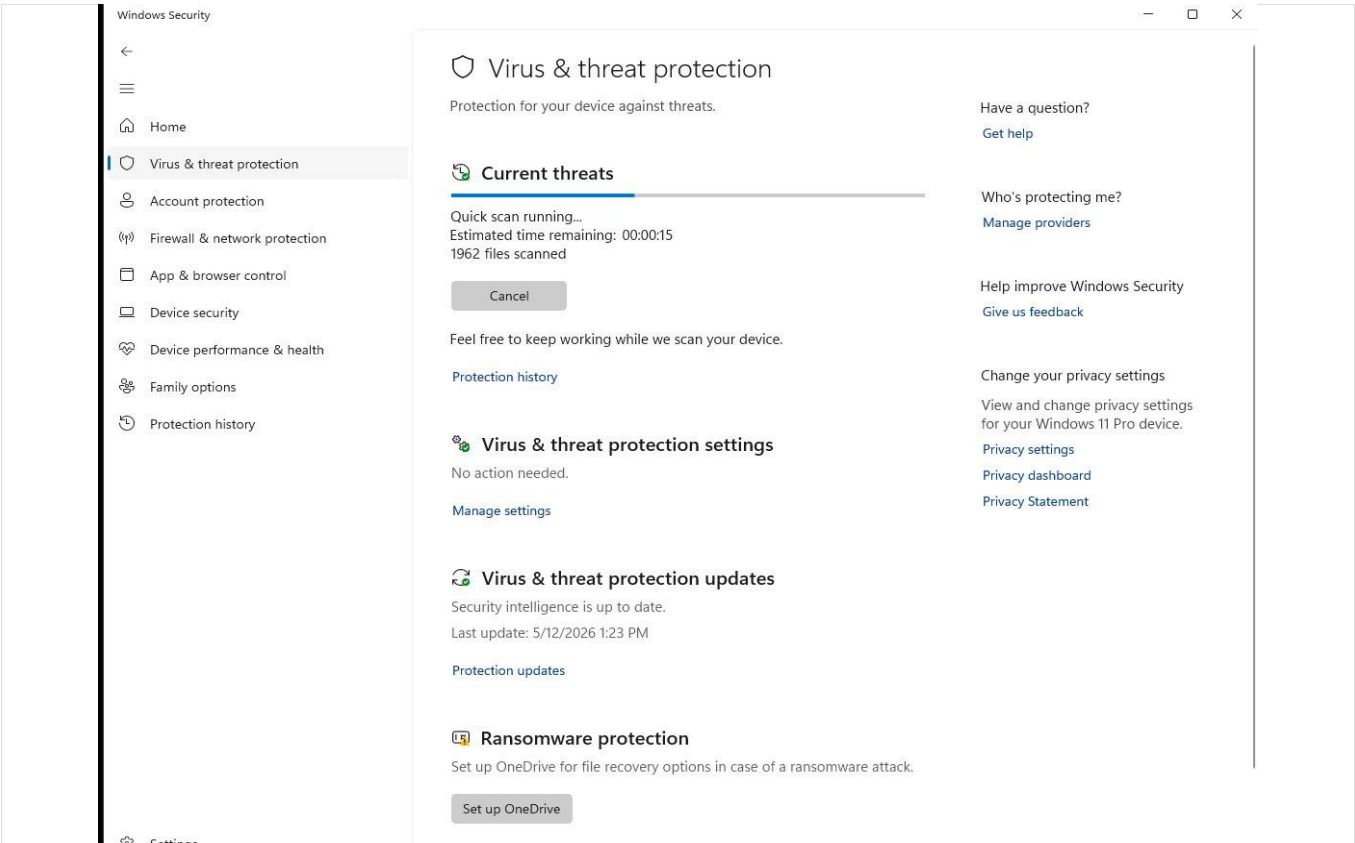


Figure 8.1: Windows Defender — Quick Scan running, 1,962 files scanned, security intelligence up to date (5/12/2026)

Defender Component	Purpose & IT Responsibility
Virus & Threat Protection	Scans for malware — ensure real-time protection is always ON, definitions updated
Firewall & Network Protection	Controls network access per profile — configure rules for allowed/blocked traffic
App & Browser Control	Blocks malicious apps and SmartScreen warnings — maintain default settings
Ransomware Protection	Controlled folder access — protect documents, enable OneDrive integration for recovery

Device Security	Core isolation, Secure Boot — verify enabled on managed endpoints
-----------------	---

8.4 Windows Firewall — Custom Rule Creation

The Windows Defender Firewall with Advanced Security console was used to create a custom inbound rule blocking ICMP (ping) traffic. Blocking ping is a common security hardening practice that prevents attackers from using network discovery tools to identify active hosts.

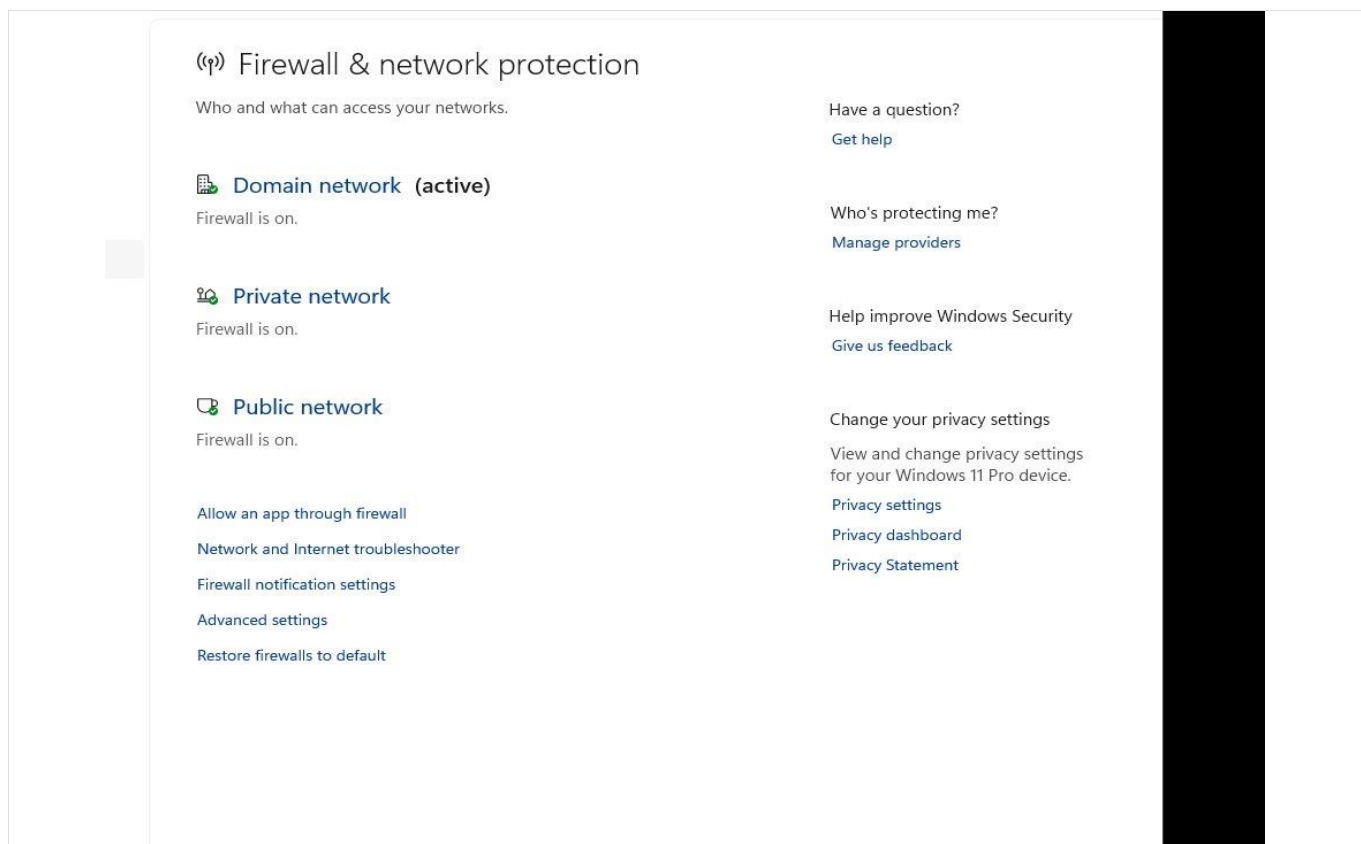


Figure 8.2: Windows Firewall showing all three profiles active — Domain (active), Private, and Public

ICMP Block Rule — Configuration Steps

32. Opened Windows Defender Firewall with Advanced Security (wf.msc)
33. Selected Inbound Rules → New Rule in the Actions pane
34. Rule type: Custom → All programs → Protocol: ICMPv4
35. Scope: Any IP address (source and destination)
36. Action: Block the connection
37. Profiles: Domain, Private, and Public — all selected

38. Name: Block ICMP → Finish

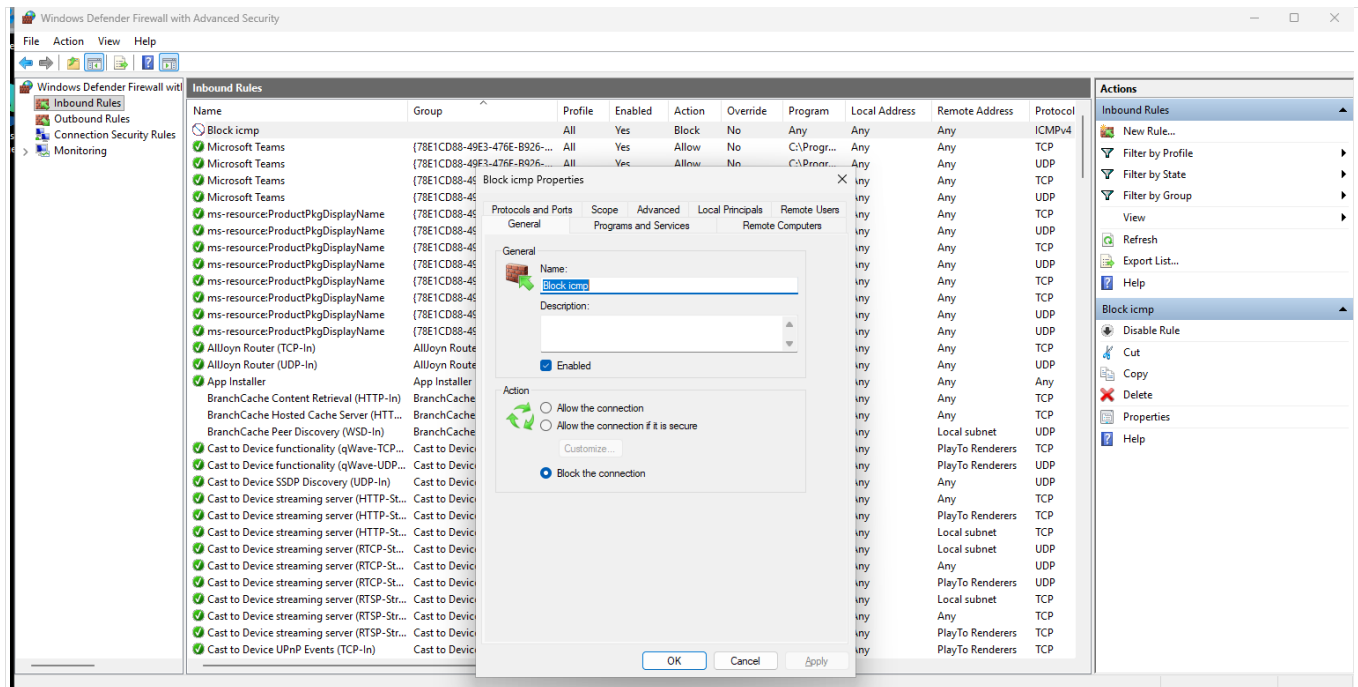


Figure 8.3: Block ICMP inbound rule — Enabled: Yes, Action: Block, Profile: All

```

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\WINDOWS\system32> ping 192.168.1.50

Pinging 192.168.1.50 with 32 bytes of data:
Request timed out.

Ping statistics for 192.168.1.50:
    Packets: Sent = 1, Received = 0, Lost = 1 (100% loss),
Control-C
PS C:\WINDOWS\system32> ^C
PS C:\WINDOWS\system32> |

```

Figure 8.4: ping 192.168.1.50 returns "Request timed out" — firewall rule successfully blocking ICMP

8.5 Common Security Threats — Reference Guide

Threat Type	Description	IT Response & Prevention
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Phishing	Fraudulent emails impersonating legitimate organisations to steal credentials or deploy malware	User security awareness training, email filtering, report suspicious emails, never click unverified links
Brute Force	Automated tool repeatedly attempts password combinations until successful	Account lockout policy (3-5 attempts), MFA enforcement, strong password complexity requirements
Ransomware	Malware encrypts user files and demands payment for decryption key	Regular verified backups, Defender real-time protection, restrict macro execution, patch promptly
Social Engineering	Manipulating users into revealing information or performing actions through psychological deception	IT staff never ask for passwords — train users to verify caller identity, escalate suspicious requests
Insider Threat	Malicious or negligent action by a current or former employee causing data loss or system damage	Principle of least privilege, audit logging, immediate account disabling on departure, access reviews
Man-in-the-Middle	Attacker intercepts communication between two parties to eavesdrop or modify data	Enforce HTTPS, require VPN on public networks, verify certificate authenticity, disable HTTP

